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| Name: | ID: | Section: |
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### Question 1 [C01] [10 Points]

Take n valid temperature readings from the user. Temperatures below 50 or above 150 are invalid. For such user inputs display: "Invalid temperature, temperature value." Continue to take user inputs until you have 'n' valid inputs. Finally, calculate a Temperature Score based on the following rules:

- ❖ Add the temperature value if,
  - value is divisible by 5
  - value is a multiple of 4
- ❖ Otherwise, subtract the temperature value.

Count the number of even digits in the Final Temperature Score. [Before counting, check if the score is negative. If it is, multiply it by -1.]

| Sample Input                                                                                                                                                                                              | Sample Output                                                                               |
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| Enter n: 5<br>Enter reading 1: 60<br>Enter reading 2: 55<br>Enter reading 3: 64<br><b>Enter reading 4: 10</b><br>Enter reading 4: 70<br>Enter reading 5: 51                                               | <b>Invalid temperature, 10.</b><br>Final temperature Score: 198<br>Number of even digits: 1 |
| Explanation: 6 temperature inputs taken as one was invalid (10).<br>Temperature score = $60 + 55 + 64 + 70 - 51 = 198$<br>The even digit of the Temperature score is 8, so the number of even digits = 1. |                                                                                             |